

Report: root-to-shoot ratio

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1 Summary

Data Exploration

- Root mass slightly more right-skewed than sprout mass
- Sprout-root-mass-ratio shows stronger peak around 0.5 - 0.7
- High correlation between sprout- and root-mass ($r = 0.76$)

Marginal Distribution

- Gamma fits data better than Normal, support with Kullback-Leibler-Divergence

Ratio Modeling

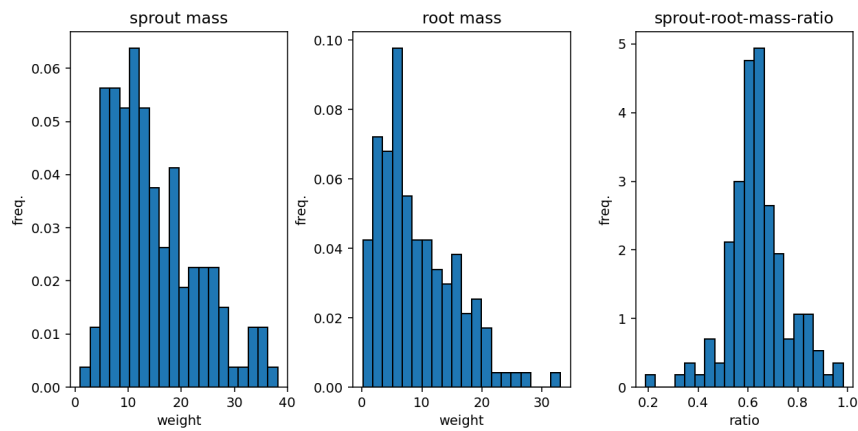
- The theoretical model doesn't match the empirical distribution of the ratios
- Supports statistical dependency of root- and sprout-mass

Interpretation

- The growth of root and sprout might be statistically dependent (with statistical power)
- Since the theoretical model underestimates the peak, one could argue that this ratio range might be an optimum for breeding
- Since I don't have much information about the data, I cannot interpret much about the biology

2 Data Exploration

Data Distribution



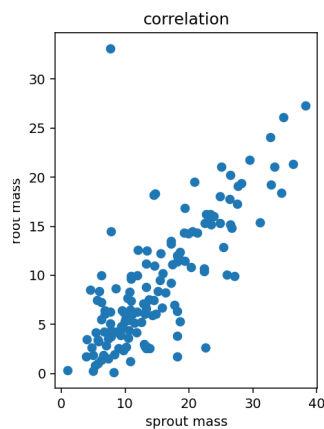
Masses

- The masses of sprouts and roots tend to be slightly right skewed (rechtsschief)
- $Weights \in \mathbb{R}_+$
- Just a normal distribution might now be appropriate, because its highly likely it will extend into the negatives and doesn't catch the skewness
- A Gamma might fits the marginal distribution better, because: $X \in \mathbb{R}_+$

Ratio

- The skweness seem to not be present here
- $Ratio \in [0, 1]$
- Intuitivly (and by their definition) a beta might fit best here

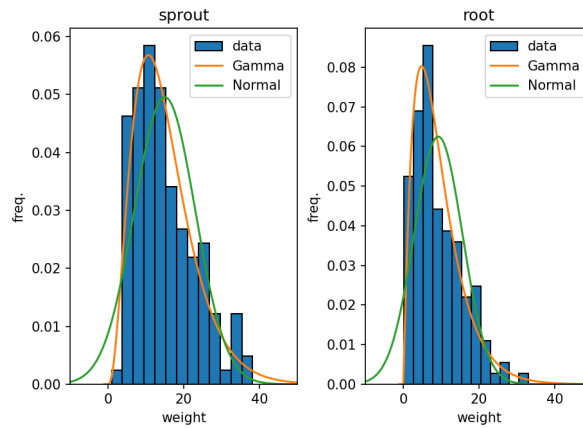
Data Correlation



- Hight correlation between the masses (=0.76)
- Implying a statistical dependency between both parameters !!!

3 Marginal Distribution

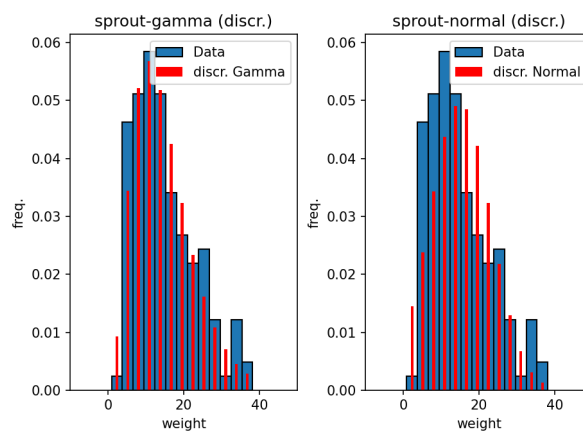
Gamma fits data better then a Normal



- Just visually I would argue that the Gamma fits X_{sprout} and X_{root} better then a Normal
- Especially, since a Gamma can model the skweness of data
- One can clearly see, that the normal extends into the negatives

Further Evaluation with Kullback-Leibler-Divergence

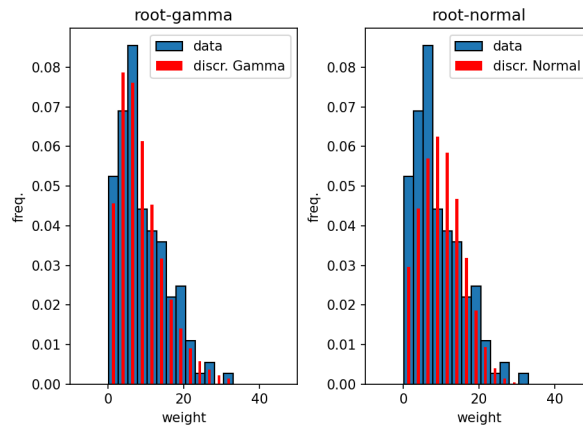
Sprout



	Gamma	Normal
KLD:	0.01999	0.05624

- Gamma has a much lower KLD

Root



KLD: Gamma Normal
 0.02861 0.06414

- Also here, Gamma has a much lower KLD

Estimated parameters from Gamma (sprout and root)

Params.	Value
a_sprout	3.4580
b_sprout	0.2310
a_root	2.0912
b_root	0.2266

- One can notice that: $\beta_{root} \approx \beta_{sprout}$
- This supports the theoretical distribution for the ratio transformation later
- But also that: $\alpha_{root} < \alpha_{sprout}$, implying that the skewness of the root distribution is stronger !

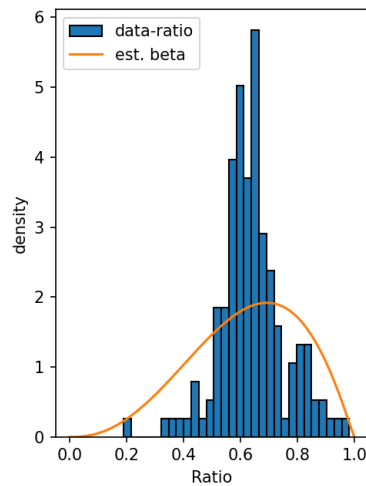
4 Ratio-Modelling

Root-To-Sprout Ratio

Theoretically

$X_1 \sim Ga(x_1 \alpha_1, \beta)$	Shown in previous section
$X_2 \sim Ga(x_2 \alpha_2, \beta)$	Shown in previous section
$Y = \frac{X_1}{X_1 + X_2}$	Transformation
$Y \sim Be(y \alpha_1, \alpha_2)$	Under the assumption that $\beta_{root} = \beta_{sprout} = \beta$ and $X_1 \perp\!\!\!\perp X_2$ (independent). This was proven in our lectures.

- I can theoretically model the ratio with just the α 's from the Marginal Gamma's



- One can clearly see that the theoretically derived beta distribution doesn't fit the empirical ratios

Interpretation

- The poor fit quality of the estimated beta distribution is an indicator that both masses are statistically dependent and further supports the correlation
- This validates, that the correlation breaks the independency assumption of the theoretical beta
- One could interpret that this peak is the result of stronger dependency (also correlation) of those seedlings masses
- Unfortunately, I don't have too much information about the data, especially I don't know if certain seedlings perform better than other (for breeding purpose)
- I could argue that this ratio range (0.5 - 0.7) might be a optimal target for breeding selection (with statistical power)